

SUMMARY

Software engineer with expertise in designing and implementing scalable Artificial Intelligence systems, relational databases, and modular software architectures. Proficient in Python, Java, Node.js, and database management, with a strong focus on object-oriented design, parallel processing, and data pipeline automation. Passionate about problem-solving through innovative and efficient solutions.

EDUCATION

Johns Hopkins University

Master of Science, Computer Science, Whiting School of Engineering

Expected Graduation: May 2026 | Artificial Intelligence Focus

Virginia Polytechnic Institute and State University, Blacksburg, VA

Bachelor of Science, Computer Science, College of Engineering

Graduated: May 2023 | Minor in Music

EXPERIENCE

Software Engineer (Contractor), Cspeed Inc. | Palo Alto, CA | Dec 2023 – March 2025

- Designed, implemented, and maintained a relational database to store and manage Silicon Photonics chip testing data, ensuring efficient data retrieval and analysis through containerization of MySQL and phpMyAdmin via Docker.
- Developed a full stack, modular, object-oriented Python software infrastructure from scratch to control Silicon Photonics chip testing environment, enabling seamless integration of test instruments (through GPIB using PyVISA) and streamlined testing workflows, including active alignment automation for stages, as well as full test automation
- Implemented a thread pool for multi-threaded execution of instrument operations, significantly reducing testing time
- Created automated pipelines for uploading chip testing data to the database, improving data collection efficiency and accuracy.
- Used JMP to perform chip testing data analytics and data visualization, providing actionable insights for improving chip performance.

Technical Intern, Cisco Systems, Inc. | San Jose, CA | May 2022 – August 2022

- Built a Node.js application using Zuora's AQUA API to enable customizable queries in MongoDB, delivering finance-related data via a Webex Chatbot.
- Configured Webex spaces to receive query results, improving accessibility to critical financial information.

Technical Intern, Cisco Systems, Inc. | San Jose, CA | May 2021 – August 2021

- Worked in a Scrum team to develop an end-to-end tax refund feature for a Webex Teams finance bot for customer needs using [Node.js](#), allowing customers to document refund requests through a chatbot interface.

Technical Intern, Cisco Systems, Inc. | San Jose, CA | May 2020 – August 2020

- Worked in Cisco DevOps, Designed and implemented a Java API for Cisco's e-commerce web application, enhancing functionality and user experience, and integrated into the CI/CD pipeline via Jenkins for continuous builds and testing.
 - Created a risk management automation tool to resolve recurring vulnerabilities in Java files, deployed across multiple Git repositories.
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RESEARCH EXPERIENCE

Researcher, Intuitive Computing Lab, Johns Hopkins University

- Applied Principal Component Analysis on windowed sensor data to uncover dominant patterns in emotional expression.
 - Visualized PCA outputs and generated explained variance plots and knee plots to determine optimal feature dimensionality reduction.
 - Results will inform downstream modeling in affective computing and human-robot interaction research.
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NOTABLE AI PROJECTS

Reinforcement Learning with Q-Learning and SARSA

- Implemented and analyzed Value Iteration, Q-Learning, and SARSA agents in a non-deterministic grid-based racing simulator, evaluating their adaptation to probabilistic transition failures and safety constraints.
- Engineered and tuned hyperparameters (learning rate, discount factor, exploration schedule) via grid search, achieving up to 60% faster convergence and improved stability in model-free algorithms across three distinct racetracks.
- Demonstrated how on-policy and off-policy methods respond to unexpected state transitions through distinct crash handling variants
- Visualized and interpreted learning curves to demonstrate trade-offs in convergence speed, variance, and sensitivity to exploration

Decision Trees for Classification and Regression

- Implemented and evaluated decision tree models with reduced error pruning for classification and regression datasets
- Analyzed pruning effects on model performance, demonstrating improved generalization in regression tasks through reduced MSE
- Preprocessed datasets by encoding categorical variables and removing non-predictive features, evaluated with 5x2 cross-validation

Neural Networks with Autoencoder Pretraining

- Developed and compared linear/logistic regression performance with feedforward neural networks and autoencoder based networks implemented from scratch across classification and regression datasets
 - Conducted systematic tuning of hidden layer dimensions and encoding fractions using 5x2 cross-validation to optimize architecture
 - Demonstrated autoencoder pretraining improves classification accuracy on nonlinear datasets / linear models excel in regression tasks
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SKILLS

- Programming Languages:** Python (PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, PyVISA), Java, Node.js, C/C++, React JS, R
- Databases:** Relational Databases (SQL), MongoDB, MySQL, phpMyAdmin, ETL, Statistics, Excel
- Tools & Frameworks:** JMP, Git, Docker, Kubernetes, AWS, REST APIs, Postman, Tableau, Power BI, Unix, Jenkins, JUnit testing, Matlab, SolidWorks
- Concepts:** Object-Oriented Design, Parallel Processing, Data Pipelines, Modular Software Architecture, Agile, Scrum, Clustering
- Machine Learning:** Anthropic Claude / CNP, TensorFlow, LLM for Regression and Classification, Decision Trees (XGBoost), Deep Learning
- Other:** Linux (Ubuntu, CentOS), Operating Systems, Data Science, DSP, Network Programming, SDLC, BitBucket, Jira, SaaS